



Name \_\_\_\_\_

## Today's big idea

If you're about to do the same thing twenty times — **don't**. Tell the computer to repeat it.

| repeat 4 | CONTROL |
|----------|---------|
| clap     |         |
| clap     |         |
| stomp    |         |

That's 12 actions written in 4 lines. Change the 4 to a 100 and you've just done 300 actions by typing two characters.

And when you want the computer to surprise you, use **pick random**.

## Words to know

| Word     | What it means                                         |
|----------|-------------------------------------------------------|
| loop     | Code that repeats                                     |
| repeat n | Do this exactly n times                               |
| forever  | Do this until the game stops. Nothing can go after it |
| random   | A number the computer picks that you can't predict    |

**Golden rule:** every **forever** loop needs a **wait** inside it somewhere. Without one, Scratch freezes and you lose your work.

## Class activity A – Dancing Sprite

| when green flag clicked | EVENTS |
|-------------------------|--------|
|-------------------------|--------|

|                  |         |
|------------------|---------|
| forever          | CONTROL |
| next costume     | LOOKS   |
| wait 0.2 seconds | CONTROL |
| move 5 steps     | MOTION  |

Change the `0.2` until the walk looks right. Add `if on edge, bounce` so it doesn't escape.

### Class activity B – Rain

|                                           |         |
|-------------------------------------------|---------|
| when green flag clicked                   | EVENTS  |
| forever                                   | CONTROL |
| go to x: (pick random -240 to 240) y: 180 | MOTION  |
| repeat 20                                 | CONTROL |
| change y by -18                           | MOTION  |
| wait 0.02 seconds                         | CONTROL |

A raindrop that appears somewhere random along the top and falls. Forever.

**Now change the numbers.** Fewer repeats + bigger step = faster. Try it.

### Your project – things fall from the sky!

Today your project starts looking like a real game.

- ☐ Add a falling object sprite that fits your theme
- ☐ Make it the right size
- ☐ Give it the falling script (same shape as Rain)
- ☐ **Tune the speed** so it's catchable but not boring
- ☐ **Save it**

*It doesn't notice your player yet — you can "catch" it and nothing happens. That's next week.*

### Homework

**1. A second thing to catch.** Add another falling sprite that looks different. (*Shortcut: right-click your first one and duplicate it — the code comes with it!*)

**2. Unpredictable speed.** Use `pick random` so each drop falls at a different speed. (*Hint: the `wait` can take a random number.*)

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## Bonus challenges

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1. Draw shapes with `repeat` + `turn`. A square is `repeat 4 [move 100, turn 90]`. What makes a triangle? A hexagon? What does `repeat 36 [move 10, turn 10]` draw?
  2. Make your falling object spin as it falls.
  3. Make a starfield — five objects falling at five different speeds.
  4. Use `repeat until` instead of `forever` so it stops after ten drops. (*Hard! You'll need a way to count, which we learn in two weeks.*)
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## If something breaks

- **Flickers at the top and nothing else?** No `wait` inside the falling loop.
  - **Falls once then stops?** The whole thing needs to be inside a `forever`.
  - **Falls upward?** `change y by 18` should be `change y by -18`. Up is positive.
  - **Scratch froze?** A `forever` with no `wait` in it. Reload and add one.
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Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.